Python Zero to Projects

Chapter 4 - Practice Set

Strings

Write each program below as its own separate Python file (for example, q1.py, q2.py, and so on) and run it to check your work.

Question 1

Write a program that stores your full name in a variable and prints the name itself, the total number of characters in it (using len()), and the name converted entirely to uppercase.

Question 2

Write a program that stores the string "Programming" in a variable and uses indexing and slicing to print: the first character, the last character, the first four characters, and the last four characters.

Question 3

Write a program that stores the string "Hello World" in a variable and prints it reversed, using slicing with a step of -1.

Question 4

Write a program that stores two strings of your choice, uses the + operator to join them into one sentence with a space in between, and then uses the * operator to print a line of 30 equals signs (=) above and below the sentence.

Question 5

Write a program that stores a sentence in a variable and asks the user to type a word using input(). Using the in and not in operators, print a message stating whether or not the typed word appears inside the sentence.

Question 6

Write a program that stores the string "Snake" in a variable and demonstrates that strings are immutable: attempt to change the first character to "C" using index assignment so that Python raises an error, then below that (commented out or explained with a print statement) show the correct way to build "Cnake" by creating a new string instead.

Question 7

Write a program that stores a sentence containing extra spaces at the beginning and end (for example "   Python is fun   ") and prints: the original sentence wrapped in square brackets, the sentence after strip() wrapped in square brackets, and the sentence converted to lowercase using lower().

Question 8

Write a program that stores a sentence of at least six words in a variable, uses split() to break it into a list of words, prints that list, prints the data type of the result using type(), and also prints how many words the sentence contains using len() on the list.

Question 9

Write a program that stores a sentence in a variable, uses replace() to swap one word in it for a different word of your choice, and uses find() to print the index at which a word that IS present in the sentence first appears, as well as the result of calling find() on a word that is NOT present in the sentence.

Question 10

Write a program that stores a person's name and age in two variables and prints a message in the format "<name> is <age> years old." three separate times: once built using the % operator, once built using the .format() method, and once built using an f-string.